

Helixorin — Commercial Stability Protocol

Protocol STAB-P-2025-40

Container Closure and Orientation

Samples are stored in the commercial container closure system. Bottles are stored upright; a subset of inverted samples is included at the initial and final timepoints to assess closure-liner interaction. Induction seal integrity is verified at each pull.

Storage Conditions

The primary (long-term ICH condition) storage condition for this protocol is $25^{\circ}\text{C} \pm 2^{\circ}\text{C}$ / $60\% \text{ RH} \pm 5\% \text{ RH}$. Accelerated studies at $40^{\circ}\text{C} \pm 2^{\circ}\text{C}$ / $75\% \text{ RH} \pm 5\% \text{ RH}$ were completed through six months during development and are not part of this protocol.

Testing Schedule

Interval (months)	0	3	6	9	12	18	24
Long-term testing	X	X	X	X	X	X	X

Deviations

The 18-month samples for batch 002 were held at ambient conditions for 16 hours during chamber maintenance before testing; trending showed no anomalous results.

Acceptance Criteria

The stability acceptance criterion applied at each interval is — Assay: 95.0–105.0% of label claim. All other registered specification tests apply unchanged.

Post-Approval Commitments

As a condition of approval, three consecutive production batches of Helixorin will be entered into the stability program, and the retest period will be re-evaluated after 36-month data are available.

Bracketing and Matrixing

In accordance with ICH Q1D, bracketing is applied: the 40 mg and 75 mg strengths are tested, and the intermediate 50 mg strength is covered because all strengths are direct scale-equivalents of a common blend in the same container closure system.

Photostability

Confirmatory photostability testing per ICH Q1B was completed during development and is not repeated in this protocol. The product is labeled for storage in the original container to protect from light.

Container Closure and Orientation

Samples are stored in the commercial container closure system. Bottles are stored upright; a subset of inverted samples is included at the initial and final timepoints to assess closure-liner interaction. Induction seal integrity is verified at each pull.

Analytical Procedures

Assay and degradation products are determined by the stability-indicating reversed-phase HPLC procedure described in the analytical methods dossier. Dissolution testing uses USP Apparatus II at 50 rpm. All procedures were validated in accordance with ICH Q2(R1) prior to protocol initiation.

Batch Selection

The protocol covers three primary batches manufactured at commercial scale using the approved process. Batch genealogy, manufacturing dates, and packaging lots are recorded in the batch appendix. All batches were released against the approved specification prior to set-down.